

```
import pandas as pd
df = pd.read_csv('/content/TeachingRatings.csv')
```

```
total = len(df)
greater_4_5 = (df['eval'] > 4.5).sum()
probability = greater_4_5 / total
```

```
print("Number of scores greater than 4.5:", greater_4_5)
print("Total observations:", total)
print("Probability:", probability)
print("Percentage:", probability * 100)
```

```
Number of scores greater than 4.5: 79
Total observations: 463
Probability: 0.17062634989200864
Percentage: 17.062634989200866
```

Formula

$P(\text{eval} > 4.5) = \text{Number of eval scores} > 4.5 / \text{Total number of observations}$

Interpretation

If, for example, the result is 0.10, it means:

There is a 10% probability that a randomly selected evaluation score from the dataset is greater than 4.5.

Q2. Probability that evaluation is greater than 3.5 and less than 4.2

The question asks:

$P(3.5 < \text{eval} < 4.2)$

```
df['eval'] > 3.5
```

```
      eval
0      True
1      True
2      True
3      True
4      True
...      ...
458    False
459     True
460    False
461    False
462     True
463 rows × 1 columns

dtype: bool
```

```
df['eval'] < 4.2
```

```
      eval
0    False
1    False
2     True
3    False
4    False
...      ...
458    True
459   False
460    True
461    True
462    True
463 rows × 1 columns

dtype: bool
```

```

between_3_5_4_2 = (
    (df['eval'] > 3.5) &
    (df['eval'] < 4.2)
).sum()

total = len(df)

probability = between_3_5_4_2 / total

print("Number of scores between 3.5 and 4.2:", between_3_5_4_2)
print("Total observations:", total)
print("Probability:", probability)
print("Percentage:", probability * 100)

```

```

Number of scores between 3.5 and 4.2: 164
Total observations: 463
Probability: 0.3542116630669546
Percentage: 35.42116630669546

```

means:

Keep an observation only if its evaluation is greater than 3.5 AND less than 4.2.

✓ Q3. Two-tailed test using a normal distribution

The coach wants to know:

Is the regional players' average score different from the historical professional mean?

```

from scipy.stats import norm

mu = 12
sigma = 5.5
n = 36
sample_mean = 10.7

# Calculate z-statistic
z = (sample_mean - mu) / (sigma / (n ** 0.5))

# Two-tailed p-value
p_value = 2 * norm.cdf(-abs(z))

print("Z-statistic:", z)
print("P-value:", p_value)

```

```

Z-statistic: -1.418181818181819
P-value: 0.15613768876786

```

Double-click (or enter) to edit

Assume the standard significance level:

$$\alpha = 0.05$$

Compare:

$$p = 0.156$$

with:

$$\alpha = 0.05$$

Since:

$$0.156 > 0.05$$

we fail to reject (H_0).

Final conclusion

The two-tailed one-sample z-test gives a z-statistic of approximately -1.418 and a p-value of approximately 0.156 . Since the p-value is greater than 0.05 , we fail to reject the null hypothesis. Therefore, there is insufficient statistical evidence to conclude that the regional players' mean points per game differs from the historical professional mean of 12 points per game.